

Mohua Akter

GitHub LinkedIn

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate specializing in data science and machine learning, with hands-on experience across RAG, LLMs, computer vision, and applied statistics. Designed and evaluated multilingual chatbot architectures with strong semantic-similarity performance across Bangla, English, and Banglish; benchmarked semantic segmentation models to 0.944 mIoU. Seeking entry-level data science, machine learning, or ML research roles.

HONORS AND AWARDS

Fully Funded Merit Scholarship — East West University	Summer 2025–Spring 2026
Dean's Scholarship — East West University	Fall 2024–Summer 2025
Academic Honors: Dean's List (Fall 2024, Fall 2025)	

EDUCATION

East West University — Dhaka, Bangladesh <i>BSc in Computer Science and Engineering</i> CGPA: 3.76/4.00 140 credits earned Relevant Coursework: Machine Learning, Artificial Intelligence, Big Data Analytics, Statistics for Data Science, Statistics and Probability, Data Structures, Algorithms, Database Systems	<i>October 2022 – August 2026</i>
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TECHNICAL SKILLS

Machine Learning / AI: PyTorch, RAG, FAISS, ChromaDB, LLMs (Gemma, Qwen, Phi-2), self-supervised learning, semantic segmentation, InsightFace, ArcFace-ResNet50
Data & Databases: SQL, MySQL, Supabase, SQLite, Big Data Analytics, Statistical Analysis
Programming: Python, C, C++ | **Web:** HTML, CSS, JavaScript, PHP, Flask
Tools and Platforms: Git, GitHub, Docker, Microsoft Excel, PowerPoint, Word, Google Sheets, Google Docs
Languages: Bangla (native), English (fluent)

PROJECTS

Multilingual University Chatbot — Capstone Designed the RAG pipeline and hybrid RAG+LLM chatbot, moving from Gemma 3 to Qwen; evaluated five Rasa, RAG+LLM, and hybrid architectures for Bangla, English, and Banglish in low-resource settings. Best architecture achieved the strongest semantic-similarity (BERTScore) results across all five designs. Supervised by Dr. Mohammad Rezwanaul Huq, EWU.	GitHub
FruitSeg30 Segmentation Benchmark — Solo Benchmarked DeepLabV3, SegFormer, and YOLOv26 on 1,969 images across 30 fruit classes; YOLOv26 achieved 0.944 mIoU, outperforming both alternatives. Published processed train/validation/test splits as public Kaggle datasets.	GitHub
EWU Admission Chatbot — Solo Built a RAG admission chatbot and fine-tuned Gemma 3 on custom JSONL data covering EWU programs, tuition, policies, and the admissions calendar. <i>Tech: Gemma 3, RAG, LLM fine-tuning.</i>	GitHub
AttendX — Smart Attendance System, Solo Built a web attendance system with real-time face recognition and role-based student, teacher, and administrator portals. <i>Tech: InsightFace, ArcFace-ResNet50, Flask, Supabase, SQLite.</i>	GitHub
Learnova — LMS / Student Engagement Platform, Solo Built an AI-powered learning platform with attendance tracking, classroom management, and analytics; deployed on Vercel. <i>Tech: PHP, XAMPP, MySQL, HTML, CSS, JavaScript.</i>	GitHub
Paris2024 Database Management System Designed and implemented a relational database system for managing Olympic event data. <i>Tech: SQL, MySQL.</i>	GitHub
University Multi-Campus Network Design — Solo Designed a multi-campus network topology and addressing scheme for a university setting.	GitHub

ONGOING RESEARCH

Ninapro DB7 EMG Self-Supervised Learning Testing and comparing self-supervised learning methods on the Ninapro DB7 EMG dataset to identify the most effective approach. <i>Tech: Python, PyTorch, SSL.</i>	<i>In progress</i>
Smart Contract Rug-Pull Detection Reviewing research on rug-pull detection in smart contracts; developing an approach centered on intent and behavior-deviation analysis.	<i>In progress</i>

EXPERIENCE

Private Tutor — Self-employed, Dhaka, Bangladesh - Sole income source while completing a full-time Computer Science and Engineering degree with a 3.76 CGPA. - Teach Physics, Chemistry, Higher Mathematics, General Mathmatics and Biology to Grades 9–12.	<i>2021–Present</i>
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LEADERSHIP AND ACTIVITIES

Event Organizer, Jatrabari Ideal School and College (2017–2019) | **Anchor/Host**, Jatrabari Ideal School and College (2017–2019)